

```
rw-r--r--. root root system_u:object_r:httpd_sys_content_t:s0 /var/
www/html/index.html
```

We can see that the default type allotted to this file (SELinux object) is `httpd_sys_content_t`. Why is it so? We did not specify the file type when creating it. Why did it take this particular type and not `tmp_t` or some other type?

By default, newly created objects and subjects take the type of their parent object and subject, respectively. Since the folder `/var/www/html/` is of the type `httpd_sys_content_t`, any new files created under it will also get the same type, unless you want it otherwise. The same applies to processes (or 'subject' in SELinux terminology).

Since SELinux is not allowing us to change the security context of our file to `httpd_t`, we will need to write a few allow rules and include them in our policy. As discussed in the earlier article, the best way is to add these allow rules to our Policy Module.

But first we need to understand why exactly this action is being disallowed. Let us turn to the log files for help:

```
[root@vbg ~]# tail /var/log/audit/audit.log
```

```
type=AVC msg=audit(1261193853.109:36): avc: denied { relabelto }
for pid=2293 comm="chcon" name="index.html" dev=sda2 ino=104970
scontext=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0:c1023
tcontext=system_u:object_r:if_t:s0 tclass=file
type=SYSCALL msg=audit(1261193853.109:39): arch=c000003e
syscall=188 success=no exit=-13 a0=1a666c0 a1=3d6ce19649
a2=1a67660 a3=1b items=0 ppid=1960 pid=2293 uid=500 uid=0
gid=500 auid=0 fsuid=0 egid=0 agid=0 tty=pts0 ses=1
comm="chcon" exe="/usr/bin/chcon" aubj=unconfined_u:unconfined_r:
unconfined_t:s0-s0:c0:c1023 key=(null)
type=USER_AUTH msg=audit(1261194789.950:40): user pid=2313
uid=500 auid=500 ses=1 aubj=unconfined_u:unconfined_r:unconfined_t:
s0-s0:c0:c1023 msg="op=PAM:unix_chkpwd acct="vbg" exe="/sbin/
unix_chkpwd" hostname=? addr=? terminal=? res=success"
```

Now we can see from the AVC Denial message in the Audit Log that 'relabelto' permission is denied to the source context `unconfined_t` (bash process—my shell) for the target context `if_t`. Going by our earlier discussion (Part 5), we could create an allow rule for the above in our `test.te` file:

```
allow unconfined_t if_t : file { relabelto };
```

But will this solve the above-mentioned issue? Will not other SELinux denials crop up? The important thing to remember is that SELinux permissive mode has been given for exactly this reason—troubleshooting. So let us take the system into permissive mode and see if any more errors occur:

```
[root@vbg ~]# getenforce
Enforcing
[root@vbg ~]# setenforce 0
[root@vbg ~]# getenforce
Permissive
```

Let us also clear the log file so that it contains only the latest errors. But we cannot afford to lose important logs. So we need to first make a copy of the current log file:

```
[root@vbg ~]# cat /var/log/audit/audit.log > \
/var/log/audit/my-audit-log-18Dec09
```

Now, let us empty the log file:

```
[root@vbg ~]# > /var/log/audit/audit.log
```

...and repeat the `chcon` command:

```
[root@vbg ~]# chcon -t if_t /var/www/html/index.html
```

Let us have a look at AVC denial messages in the new log file:

```
[root@vbg ~]# cat /var/log/audit/audit.log | grep -i deni
type=AVC msg=audit(1261195413.081:42): avc: denied { relabelto }
for pid=2328 comm="chcon" name="index.html" dev=sda2 ino=104970
scontext=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0:c1023
tcontext=system_u:object_r:if_t:s0 tclass=file
type=AVC msg=audit(1261195413.081:42): avc: denied { associate }
for pid=2328 comm="chcon" name="index.html" dev=sda2 ino=104970
scontext=system_u:object_r:if_t:s0 tcontext=system_u:object_r:fs_t:s0
tclass=filesystem
```

We can see that two actions are disallowed. Let us write the allow rules for them:

```
allow unconfined_t if_t : file { relabelto }
allow if_t fs_t : filesystem { associate };
```

Let us update our module policy file, `test.te`:

```
[vbg@vbg test-SELinux]$ vim test.te
policy_module(test, 1.1)
```

```
type if_t;
```

```
allow unconfined_t if_t : file { relabelto };
allow if_t fs_t : filesystem { associate };
```

Save and close. (Do note that we have now increased the version number to 1.1.)

Compile the `test.te` source file by running the `make` command:

```
[vbg@vbg test-SELinux]$ make test.pp
```